

Project Title
Delivery Route Optimizer
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Delivery Route Optimizer** is an efficient solution designed to find the best possible routes for delivering packages to multiple locations while minimizing travel distance and delivery time. In modern delivery and logistics systems, selecting an efficient route is important for reducing fuel consumption, transportation costs, and delays. This project represents delivery locations as nodes and the roads connecting them as edges in a graph. The system uses suitable data structures and graph algorithms to analyse different routes and identify the most efficient path between locations. Algorithms such as **Dijkstra's algorithm** can be used to find the shortest path based on distance or travel cost, while appropriate data structures such as **graphs, adjacency lists, and priority queues** help in storing and processing the route information efficiently. The system can handle multiple delivery locations and determine an optimized sequence of routes for the delivery vehicle. By reducing unnecessary travel and selecting shorter paths, the proposed solution improves delivery efficiency and saves time and resources. The project demonstrates the practical application of data structures and graph algorithms in solving real-world transportation and logistics problems. Overall, the Delivery Route Optimizer provides a simple and efficient approach to route planning and helps delivery services achieve faster and more cost-effective package transportation.

Signature of the Guide